13. That was the calculation just for one element in the output matrix, element $C_{0,0}$. We do this same process for all elements in C and we get all the values. Of course, we never do this by hand, but this is what is happening under the hood when we do a tensor contraction, and this process generalizes to tensors of higher order than just matrices.

Most of the time you will see Einstein notation written without the summation symbol, where it is assumed we sum over the shared index. That is, rather than explicitly writing $C_{i,k} = \Sigma A_{i,j} \times B_{j,k}$, we often just write $C_{i,k} = A_{i,j}B_{j,k}$ and omit the summation.

Einstein notation can also easily represent a batch matrix multiplication in which we have two collections of matrices and we want to multiply the first two matrices together, the second two together, etc., until we get a new collection of multiplied matrices.

This is the Einsum equation for batch matrix multiplication,

$$C_{b,i,k} = \sum_{j} A_{b,i,j} B_{b,j,k}$$

where the *b* dimension is the batch dimension and we just contract over the shared *j* dimension. We will use Einsum notation to do batch matrix multiplication, but we can also use it to contract over multiple indices at once when using higher-order tensors than matrices.

In listing 10.3 we used `A = torch.einsum('bfe,bge->bfg',Q,K)` to compute batched matrix multiplication of the *Q* and *K* matrices. Einsum accepts a string that contains the instructions for which indices to contract over, and then the tensors that will be contracted. The string `'bfe,bge->bfg'` associated with tensors *Q* and *K* means that *Q* is a tensor with three dimensions labeled, `bfe`, and *K* is a tensor with three dimensions labeled, `bge`, and that we want to contract these tensors to get an output tensor with three dimensions labeled, `bfg`. We can only contract over dimensions that are the same size and are labeled the same, so in this